

Title

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Day Month Year

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Apparently this is a good chapter...
- Chapter ??: Passed first drafting. Very Minor corrections to be made.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and bryan comments
- Chapter ??: Not written, vaguely planned
- Chapter ??: Significant amounts of thought has already gone into this.
- Chapter ??: Not started. Will write last/almost last

Chapter 1

Experimental work on BHD chapter

An interferometer converts a change in optical path length into a change in intensity of light. One of the fundamental limits to how precisely the path light takes within an interferometer is the shot noise level of the light. In other words, as shot noise manifests as intensity fluctuations of the light, this is indistinguishable from fluctuations in the differential phase acquired by two interfering beams. In this chapter, the calibration between power measured on a photodiode and the displacement noise of a mach-zehnder interferometer is demonstrated. The results from this experiment serve as an analogy to a BHD readout schemes: as long as the the local oscillators power is much greater than the signal beams power, the shot noise limit of an interferometer with a BHD readout depends only on the power in the signal beam.

1.1 ‘Theory’

The power of the light on each photodiode, as the reflectivity, R and transitivity, T , of the beamsplitter were measured to both be 0.5 ± 0.01 , is

$$P_A = \frac{P_{\text{LO}}}{2} + \frac{P_{\text{sig}}}{2} + \sqrt{P_{\text{sig}}P_{\text{LO}}} \cos(\phi), \quad (1.1)$$

$$P_B = \frac{P_{\text{LO}}}{2} + \frac{P_{\text{sig}}}{2} + \sqrt{P_{\text{sig}}P_{\text{LO}}} \cos(\phi + \pi), \quad (1.2)$$

where P_{sig} and P_{LO} are the powers of the local oscillator and signal beams and ϕ is the phase difference between the local oscillator and signal. Subtracting the two photodiode signals gives

$$S = P_B - P_A = 2\sqrt{P_{\text{sig}}P_{\text{LO}}} \cos(\phi). \quad (1.3)$$

This signal is analogous to the one obtained from a balanced homodyne readout attached to a Michelson interferometer. In another interferometer configuration whose transfer function is $H(f)$, E_{sig} would be replaced by $E_{\text{sig}}/H(f)$.

The relationship between optical path length change, Δx and a change in power, ΔP , measured by subtracting the Mach-Zehnder’s photodiode signals is

$$\Delta z = \frac{\lambda}{2\pi} \frac{1}{2\sqrt{P_{\text{sig}}P_{\text{LO}}}} \Delta S. \quad (1.4)$$

The shot noise of a beam with power P is $\Delta S = \sqrt{2\hbar\omega_{\text{EM}}P}$. The level of shot noise comes from the total power on each photodiode. If the signal beam’s power is much

lower than the local oscillator's power, then the shot noise can be expressed as

$$\Delta z = \sqrt{\frac{2hc\lambda}{16\pi^2 P_{\text{sig}}}} \quad (1.5)$$

1.2 Apparatus

The in-air mach-zehnder is shown in Figure 1.2 and Figure 1.3. It was constructed from highly stable rigid mounts to reduce the amount of mount-vibration noise. The MZ was mounted on a thick aluminium base-plate which was isolated from the optical table by rubber dampeners. The MZ was acoustically shielded with a plastic box of a few mm thickness. As the beam size within the MZ was sufficiently small, the light power in one arm was attenuated with an ND filter as surface roughness of the ND filter did not matter. One of the corner mirrors in the MZ was glued to a PZT to allow for the length of one of the paths to be controlled. The light that emerged from the MZ was measured with the photodiode circuit shown in Appendix ???. The response of this photodiode circuit is flat at all frequencies of interest and is 390 V/W at DC.

A half wave-plate and polarised beam splitter were used to control the power entering the MZ and the polarisation of the light entering the MZ. The input laser power was monitored with a photodiode. A microphone was used to monitor the environmental noise sources.

The photodiode signals were recorded with CDS (see Appendix ??). To send signals long distances across the lab, differential send-receives circuits, constructed with the THAT range of IC, were used to remove common-mode noise that may

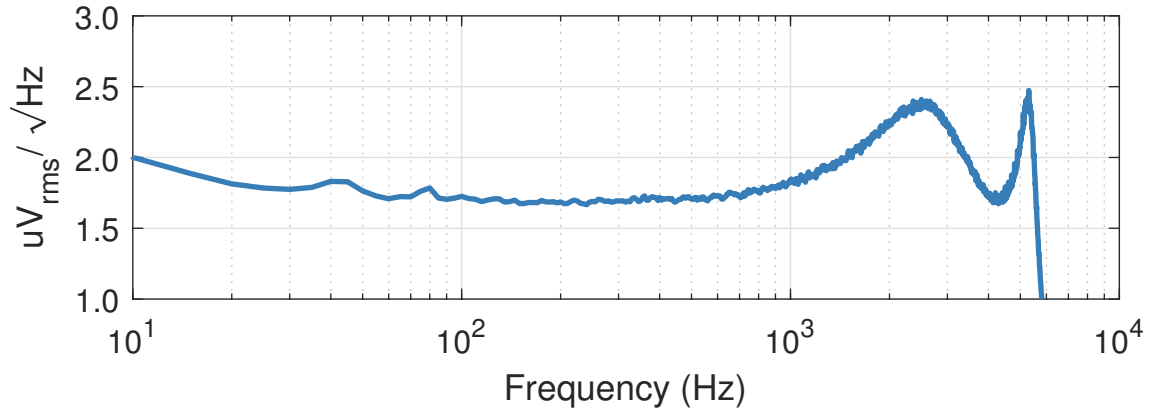


Figure 1.1

be picked up along cable. The overall factor which converts the recorded digital signal into a voltage is determined by the number of bits used in the ADC (16), the peak-to-peak voltage that the ADC can record, 20 V, and the factor of 2 that comes from the differential send-receive circuit: the factor is therefore $20/(2 \cdot 65536)$.

To keep the measured signal above the digitisation noise inherent to the analogue to digital converters used in CDS, whitening filters are required. This is a circuit which boosts the AC components of a signal while keeping the DC level of the signal small enough as to not saturate the ADC. A dewhitening filter is a digital filter whose transfer function is the reciprocal of the whitening filter's. In addition to this, CDS features an anti-aliasing filter which will effect measurements that are made near the Nyquist frequency of the ADC. This response of the AA filter can be seen in figure 1.1. CDS noise is shown in figure 1.1. For this photodiode circuit, the level of shot noise due to 20 mW of light is $\sim 3 \times 10^{-8} \text{ V}_{\text{rms}}/\sqrt{\text{Hz}}$ and so the whitening filter needs to boost the photodiode signal by a factor of 100.

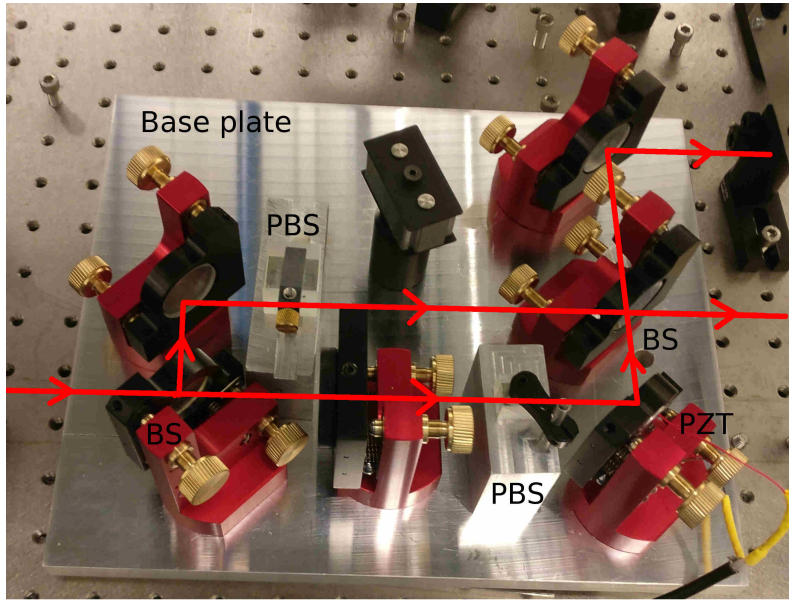


Figure 1.2: Photograph of the Mach-Zehnder base plate. The polarising beam splitters were part of an earlier design; they were removed when the measurements with this apparatus were made.

To keep the path length difference between the two arms fixed, a control signal was applied to the PZT. This was implemented with the circuit shown in Figure ?? . The servo shaping was a low pass filter with a corner frequency of 1 Hz. The PZT acts as a low pass filter with a corner frequency that is determined by its capacitance. To prevent oscillations due to the servo being unstable, the overall gain of the servo was controlled with a potentiometer.

1.2.1 Calibration of In-Air Mach-Zehnder

The shot noise level can be found by summing P_A and P_B . The power was measured to be 11 mW, so shot noise is... ?? (labbook post with data).

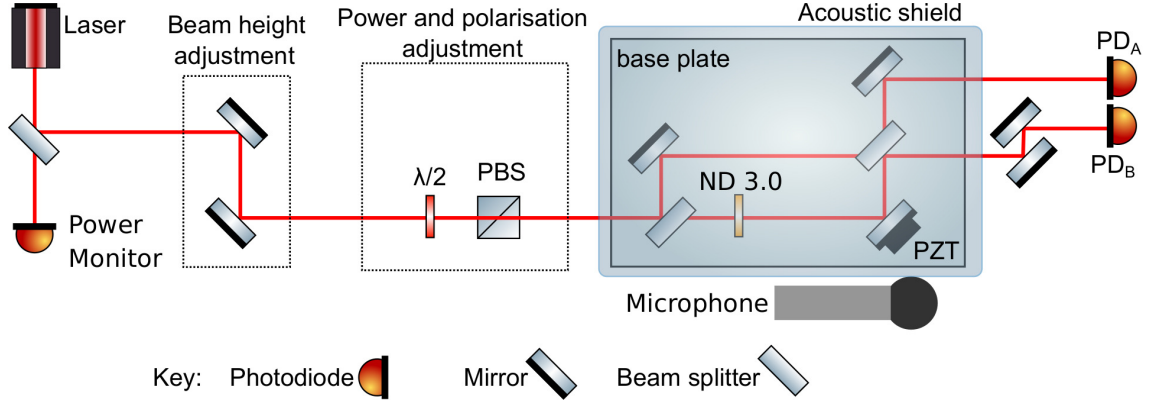


Figure 1.3: Sketch of the optical layout. The laser was a farady-isolated NPRO beam. The power of the laser was monitored using a photodiode immediately after the laser. Beam power and polarisation control was situated immediately upstream of the Mach-zehnder. The . A microphone and a humidity sensor where placed near the MZ.

The Mach-Zehnder can be calibrated by measuring P_{sig} and P_{LO} . The power in the LO path was measured to be 20 mW (LO and Sig power measurement labbook post), and the power in the lower path before the beam passed through the ND filter was measured to be 17 mW. After the ND filter (ND filter labbook post), the power would be 0.1 mW. This is shown in figure 1.4.

Alternatively, the Mach-Zehnder can be calibrated by measuring $\Delta S/\Delta z$ by applying a ramp signal to the PZT. The ramp signal and the photodiode signals were recorded on an oscilloscope. The photodiode signals can were then plotted against the measured ramp signal, and data was fit with Equation 1.3. This is shown in figure 1.4.

Parameter	Value
Nyquist Frequency	32768 Hz
Frequency Resolution	10 Hz
Effective Noise bandwidth	15 Hz
Number of Averages	459

Table 1.1: A Hanning window was used to average the data and generate the amplitude spectral density.

Although there is a 20% discrepancy between the two calibrations, both methods of calibrating the response of the Mach-Zehnder give consistent results within margin of error determined by the quantisation noise associated with the oscilloscope and the uncertainty of the signal beam's power.

1.3 Shot noise measurement

The MZ was locked and the noise measured by both photodiodes was recorded via CDS. The measured signal corresponds to the level of shot noise predicted by equation 1.5 at frequencies where classical noise sources no longer dominate the signal. To demonstrate that the signal depends on the differential path length of the MZ, a modulation at 6033 Hz was applied to the PZT. This can be seen in the spectrum shown in figure 1.5. By analogy between PZT motion and noise level, for a 100 Hz measurement bandwidth, differential path length changes smaller than $5 \times 10^{-14} \text{ m}_{\text{rms}}$ cannot be measured as the measurement will be limited by the power of the signal beam.

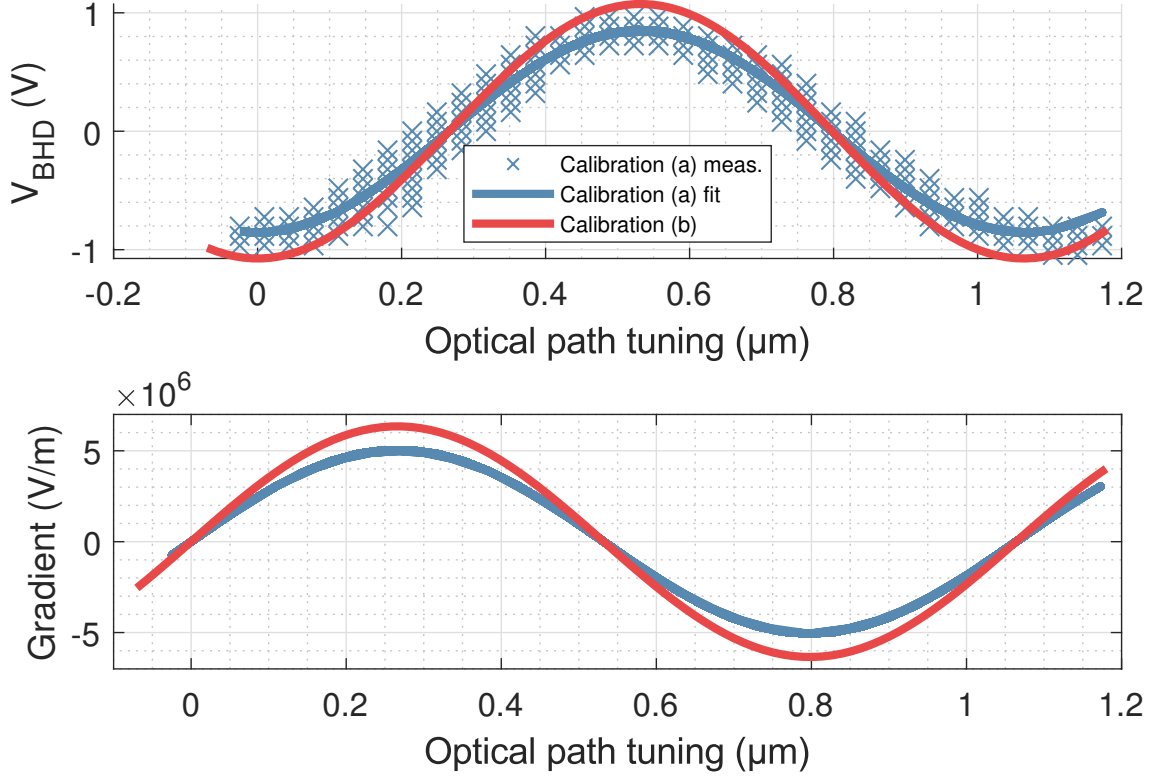


Figure 1.4: There are two ways of calibrating the photodiode signal that will be generated by a change in optical path length between the two paths in the Mach-Zehnder. Method (a) is to apply a ramp signal to the PZT while simultaneously measuring the photodiode signals. The measurement of this is shown by the blue crosses, and the fit to this data is shown by the blue line. Method (b) is to measure the power in the local oscillator and signal beams when they are not interfering. From this, Equation 1.3 can be used to get the response of the Mach-Zehnder. This is shown by the red line. The blue and red data are independent of each other, within 20% of each other and are in agreement with each other. The gradient of V_{BHD} is used to get the conversion factor for beam power to path length change.

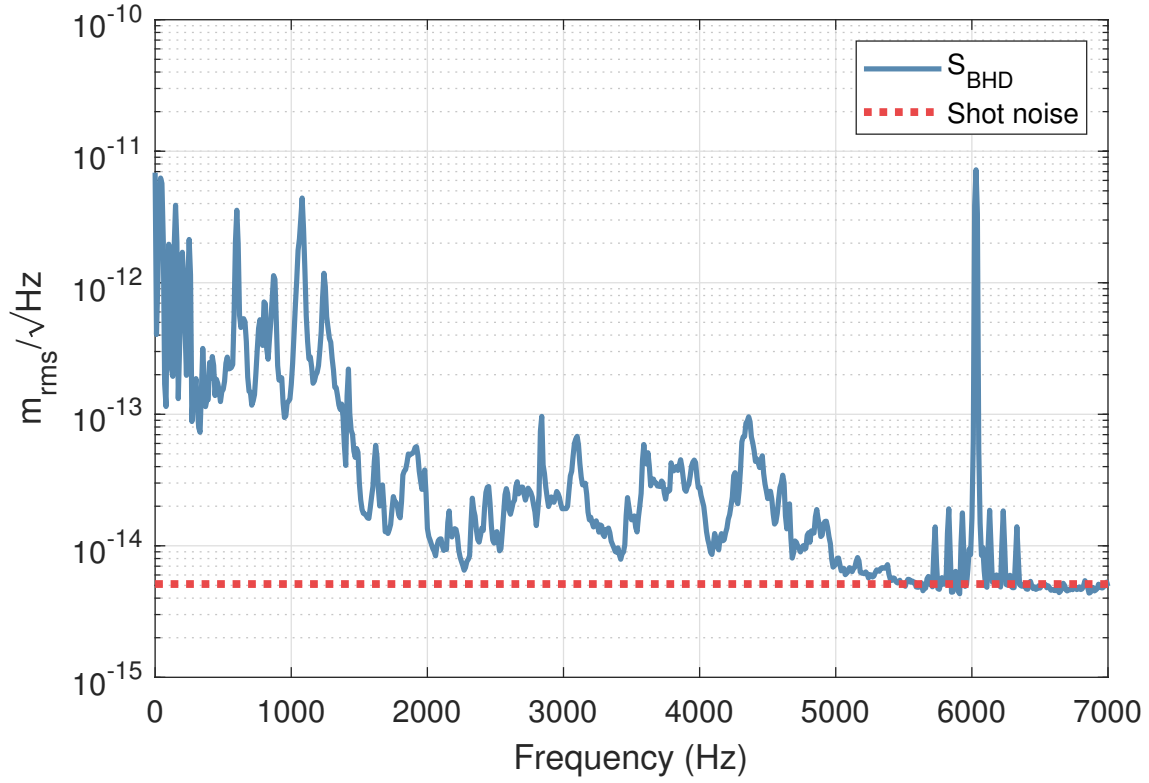


Figure 1.5: Shot noise was measured at frequencies above 5500 Hz. Below 5500 Hz, acoustic noise limited the MZ sensitivity. The peak at 6033 Hz was injected to demonstrate the effect of a differential arm length modulation.

1.4 In-Vacuum work

To measure shot noise at frequencies around 100 Hz, the frequency band of interest in a gravitational wave detector, the acoustic and seismic noise which shake the mirrors need to be reduced. This can be achieved by using suspended optics and performing the experiment in-vacuum. There was an opportunity to perform this experiment in-vacuum with left-over apparatus from a previous experiment. This proved challenging for as the interferometer was hard to lock due to large amounts of seismic motion coupling to the differential arm length. Ultimately, scattered light interfering with the differential arm signal meant that shot noise could not be measured at any frequency. However, this apparatus could be used to verify that an angular misalignment, θ , will cause the signal, S , to drop from its maximum value, S_0 , as

$$S_{\text{sig}} = S_0 e^{-(\theta/\theta_c)^2}, \quad (1.6)$$

where $\theta_c = \sqrt{2}\lambda_{\text{EM}}/\pi w_0$ is the characteristic angle of the a beam of waist w_0 .

1.4.1 Apparatus

An NPRO was used to The suspensions used were double pendulums of length and mass ??, resulting in them having longitudinal resonances ~ 1 [?].

1.4.2 Servo calibration

The differential arm length needed controlling below the suspension main resonance, ~ 1 Hz as the optics moved on the order of one wavelength at these frequencies. Due to the design of the of the balanced homodyne readout, there are lots of suspension-

related modes that couple to the longitudinal signal – these are around 30 Hz. This is made extra bad by the optical bench – this moved around a large amount due to poor seismic isolation made the optical table move around more than the floor at these frequencies. This combination of factors meant that the differential arm length noise was at a similar level at $\sim 30\text{ Hz}$ as it was at $\sim 1\text{ Hz}$. The suspended mirror was actuated on with a magnet+coil to lock the MZ arm length, this meant that the actuators strength fell as $1/f^2$ above the suspensions resonance, meaning that the coil would need to drive a signal that was provide a large signal at 1 Hz, as well as one that was 1000 times greater at $\sim 30\text{ Hz}$. The magnet+coil strength was limited by geometry (size of the mirror, size of the magnet, size of the coil, amount of current that the coil could have etc) and so was not strong enough to do this. This meant that the unity gain point needed to be between 1 and 10 Hz. The servo was design to: achieve a unity gain point just above $\sim 1\text{ Hz}$ so that the main suspension resonance could be prevented from unlocking the MZ, and filtering out the $\sim 30\text{ Hz}$ modes from making using up the dynamic range of the servo.

The servo consists of x logical blocks: the digital filtering, the sends/receive circuits, the ,the coil-driven suspended mirror, the photodiodes

1.4.3 Misalignment section

1.5 Conclusion

this experiment nearly ruined my PhD!

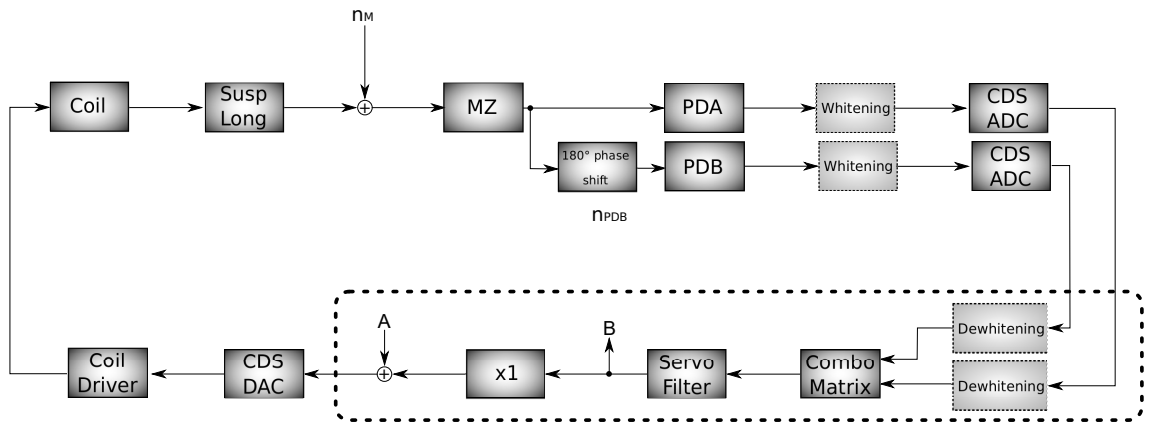


Figure 1.6

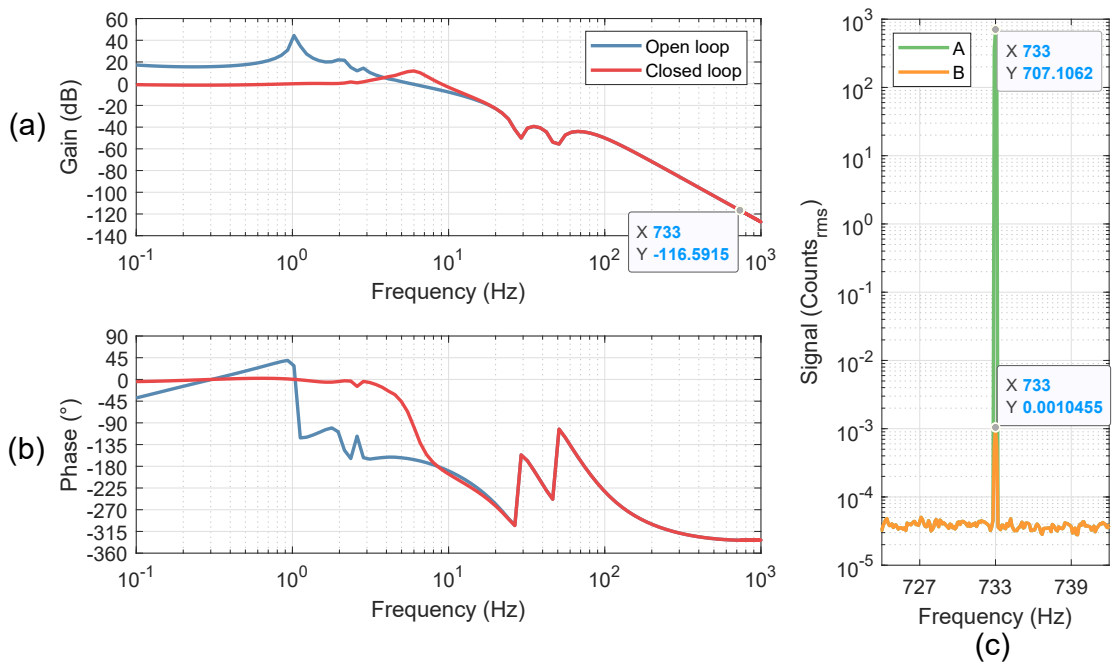


Figure 1.7

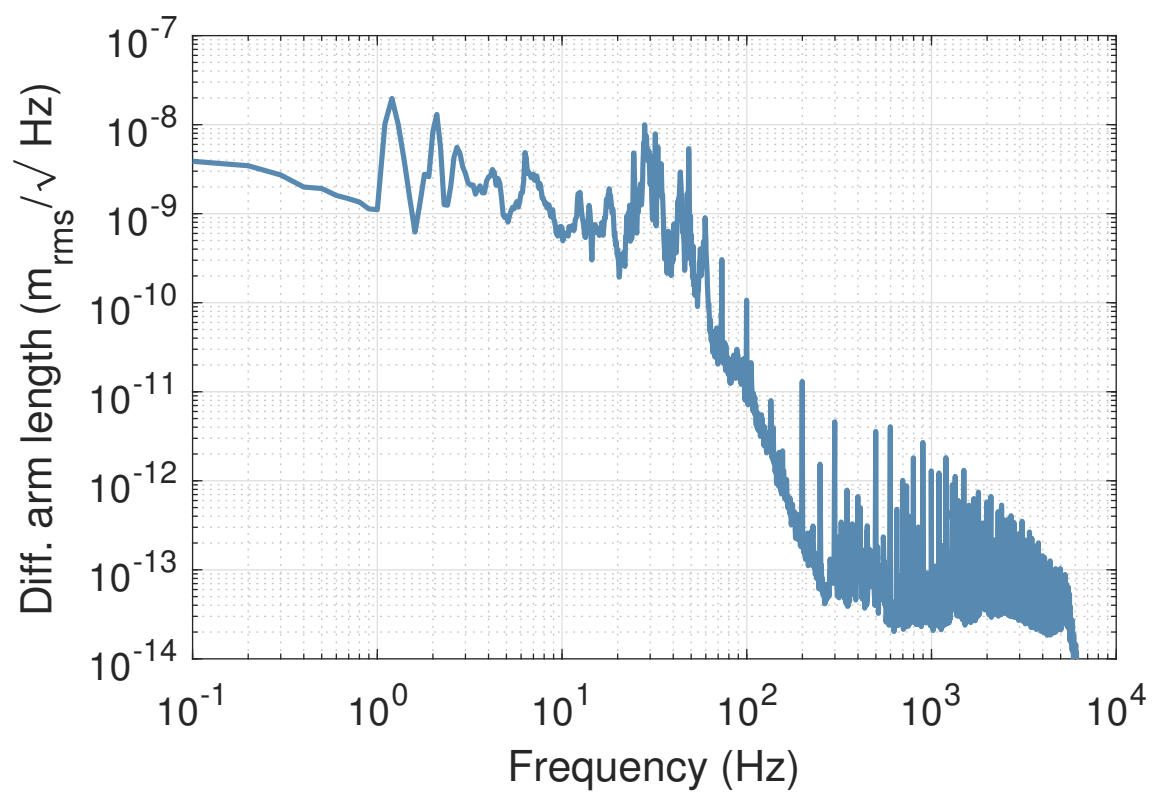


Figure 1.8

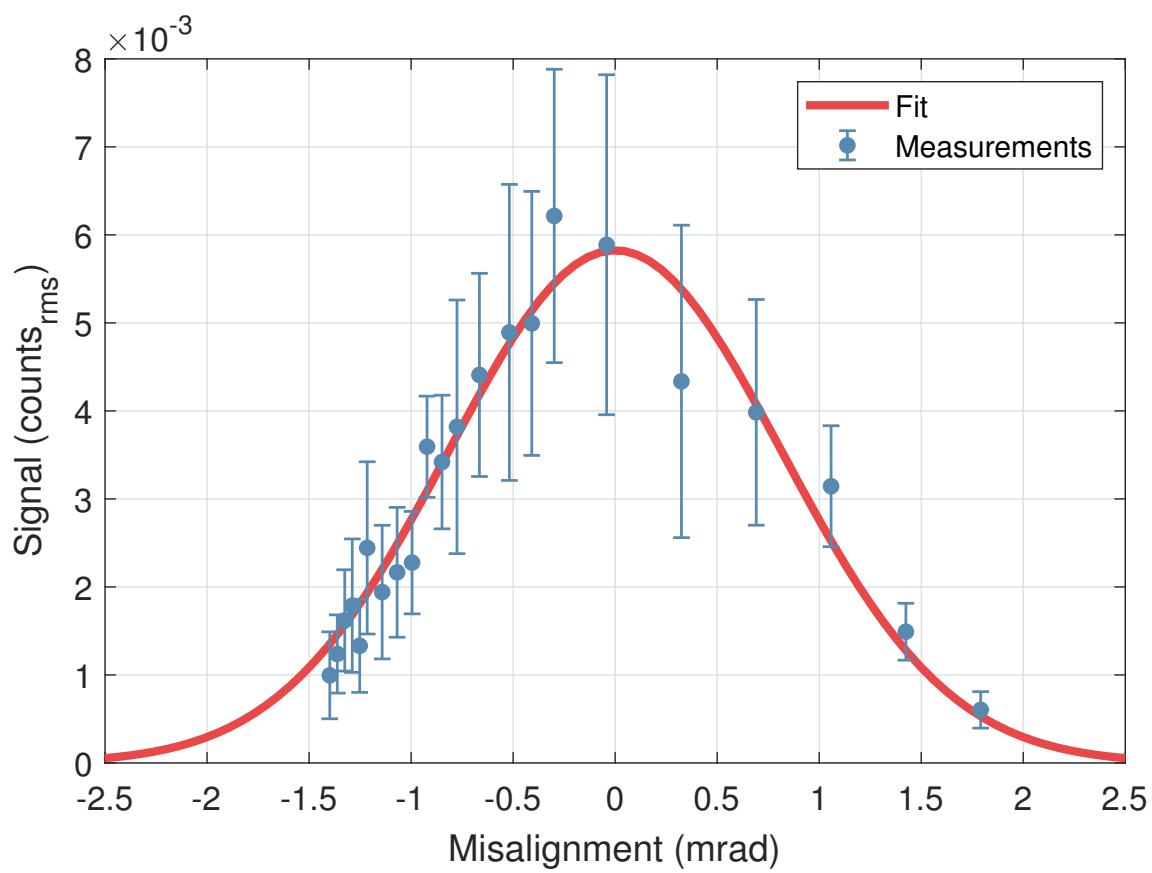


Figure 1.9

Bibliography